Do Now · retrieval

Recursion: write and debug algorithms

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

SumTo(n) returns 1 when n=1; otherwise it returns n+SumTo(n-1). For SumTo(4), identify the call that returns first and the call that returns last.

Q2

Why must a recursive function contain a stopping condition?

Q3

A loop repeatedly halves n starting at 40 while n > 1 using integer division. List the next three n values.

Q4

A function PowerTwo(n) returns 1 when n=0, otherwise 2*PowerTwo(n-1). What does PowerTwo(3) return?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Find the smallest problem — A positive single-digit number needs one digit. So n < 10 is our base case.

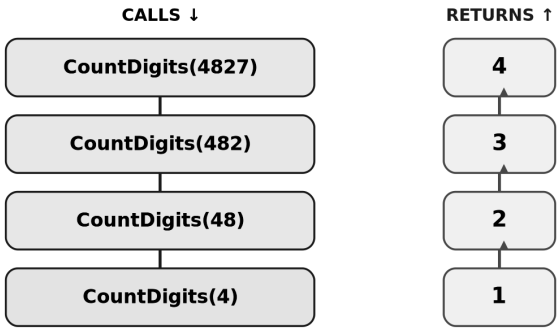
Core idea 2

Make progress — For n ≥ 10, integer division by 10 makes the number shorter: 4827 → 482 → 48 → 4.

Core idea 3

Count on the way back — 4 returns 1. Then 48 returns 2; 482 returns 3; 4827 returns 4.

Key visual · complete concept



New Learning · applying the ideas

Vocabulary · use the correct technical terms

termination — Reaching a condition where the algorithm stops.

recursive step — Processing part of a problem before calling the function on a simpler input.

precondition — An assumption or input rule required for an algorithm.

stack overflow — A failure that can occur when too many active calls consume stack space.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

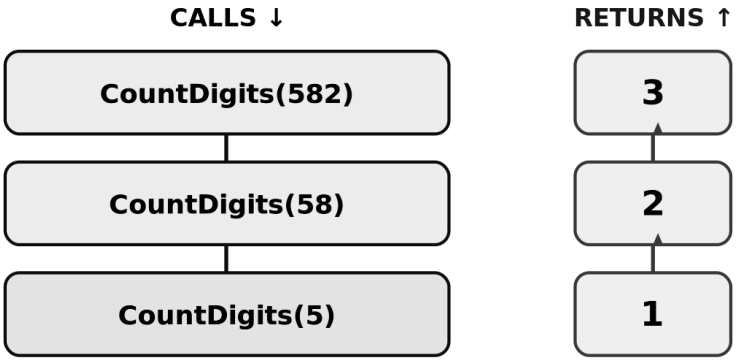
Problem

Model CountDigits(582). Why is integer division essential?

Follow the worked method

1. Calls — 582 → 58 → 5. Each integer division by 10 discards the last digit.
2. Base case — 5 is a single digit, so CountDigits(5) returns 1.
3. Returns — CountDigits(58)=2; CountDigits(582)=3. The original number has 3 digits.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Calls
9031, 903, 90, 9.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Trace CountDigits(67): write the call sequence and final value.

Question 2

A pupil writes return CountDigits(n) + 1; for the recursive step. Correct the line and explain the error.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Extend CountDigits for n = 0 and negative integers without overflowing on the minimum C# int value.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

An algorithm is recursive whenever it uses a loop.

Use key vocabulary from memory

1. Explain termination in your own words.

2. Explain recursive step in your own words.

3. Explain precondition in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.